CSE 490/590 Computer Architecture, Spring 2026
Appendix 1 – Accessing Xilinx Vivado

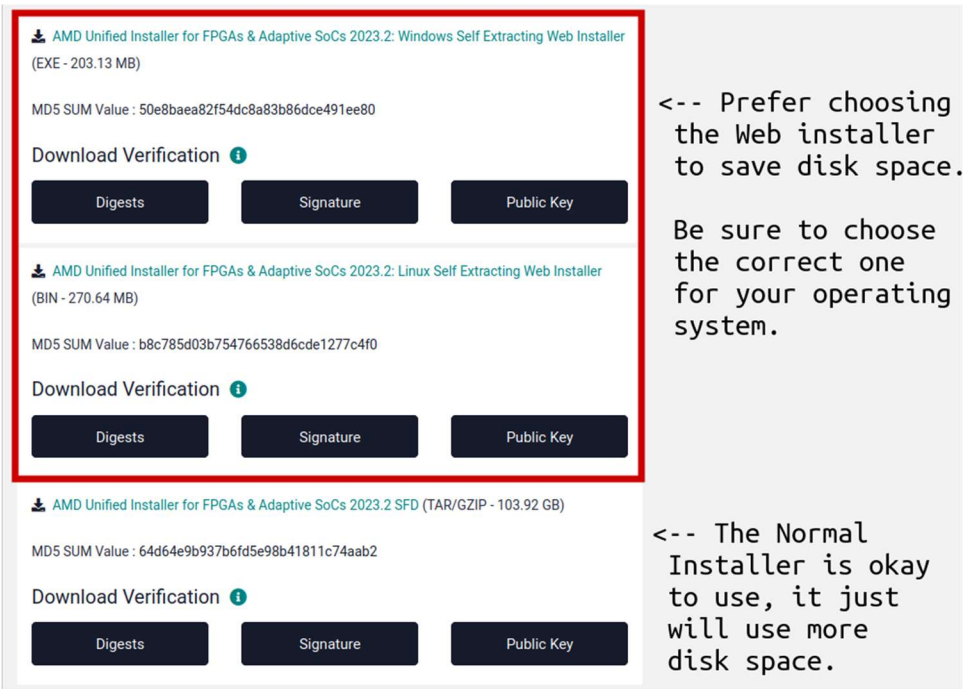
You have two options for accessing and using Vivado: either locally installing Vivado on your computer or remotely accessing Vivado on from the SENS/CSE servers.

1 – Steps for Local Installation

WARNING: Installing the Vivado Design Suite locally onto your personal computer will take several hours for the initial installation to run, and will require about 70 GB of storage to install (less after completing the installation). Having Vivado installed on your own computer will be convenient, but if you do not have enough space available, it is highly recommended to access Vivado remotely instead.

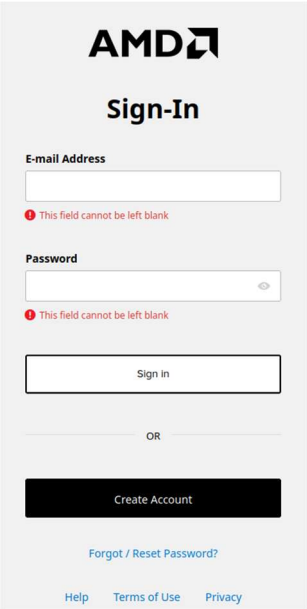
Step 1.1: Download the installer from the website

Goto https://www.xilinx.com/support/download.html and download version 2025.2 from the website. Using the web installer is preferred since they will use less disk space.



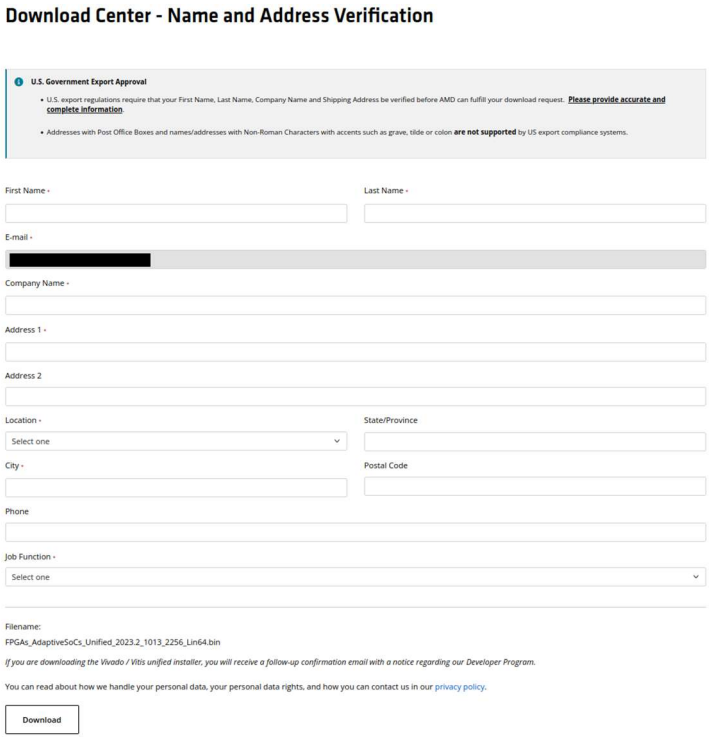
Step 1.2: Create (or login to) an AMD account

After clicking on the download option of your choice, you will be prompted to create/login to an AMD account.



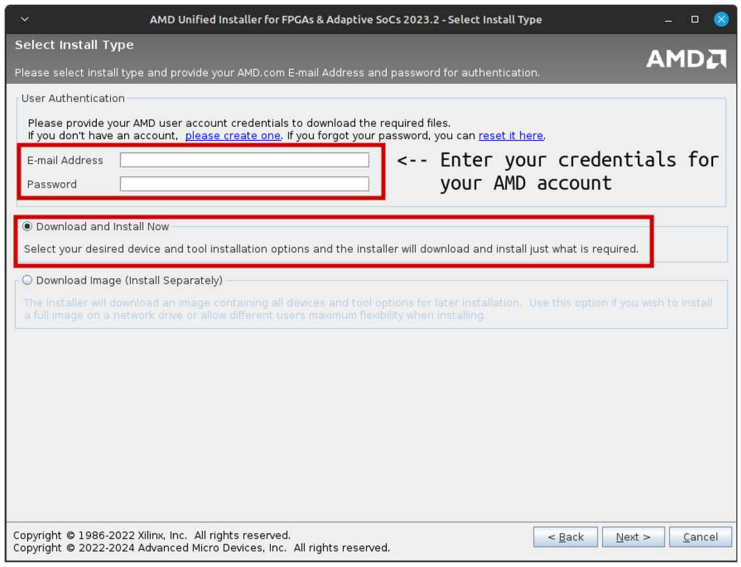
Step 1.3: Fill Out United States Export Approval Form

Before you are allowed to download Vivado, you are required to provide some personal information due to U.S. export regulations.

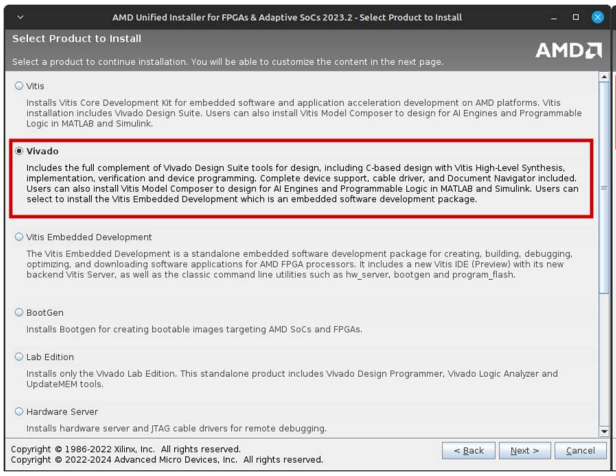


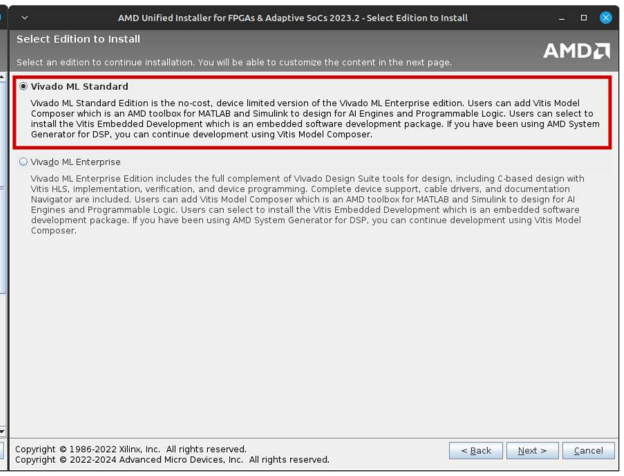
Step 1.4: Run the Installer (Web Installer)

Run the installer, you will be asked to enter your email address and password for your AMD account. Select “Download and Install Now”.

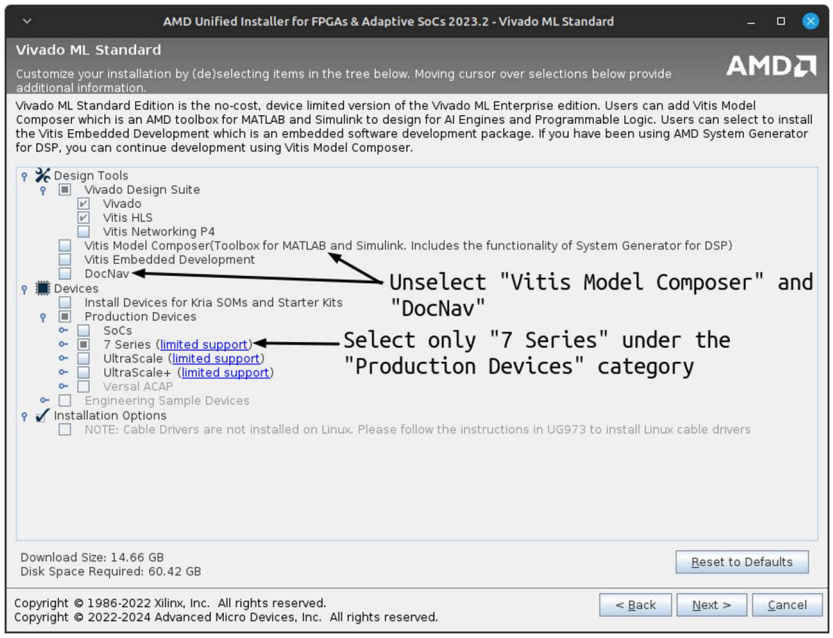


Select “Vivado” and then “Vivado ML Standard”.

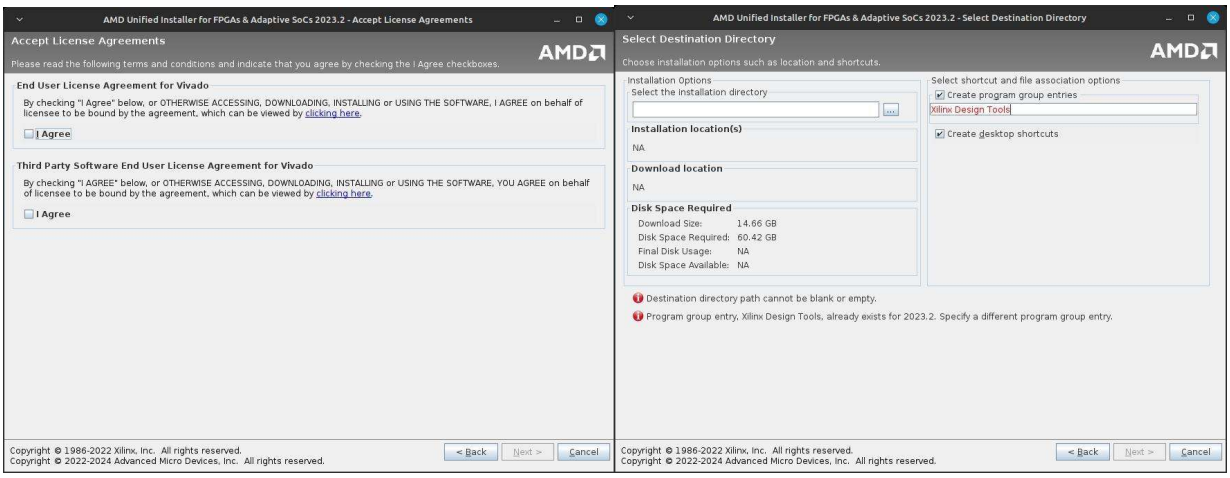




Next unselect all options except “Vivado”, “Vitis HLS”, and “7 Series”.



Finally, agree to the License Agreements and set options for your installation.

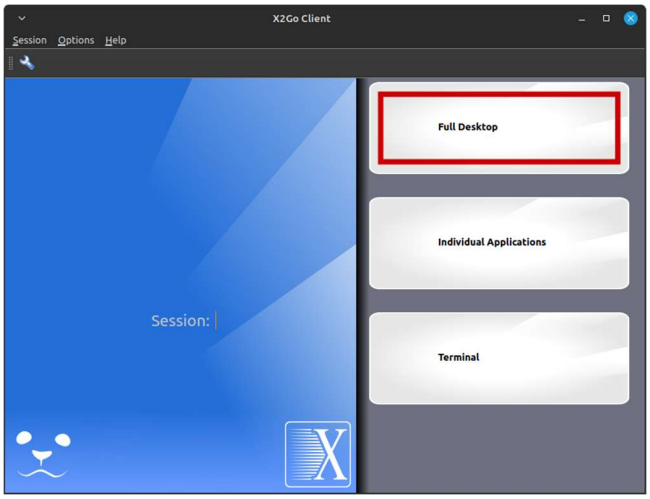


2 – Steps for Remote Access from SENS/CSE Servers

Step 2.1: Connect to SENS servers using X2go

1. Read the instructions from the SENS X2go information page and follow the instructions for installing X2go on your personal machine.
2. Make sure that you are connected to UB's Local Network by either connecting to UB's WiFi or ethernet while on campus, OR connecting to UB's VPN (https://www.buffalo.edu/ubit/service-guides/connecting/vpn.html) when off campus.
3. Under the "Connecting" section of the SENS X2go information page follow the directions for your specific operating system for connecting to the SENS servers.
4. Enter your UBIT name and password.

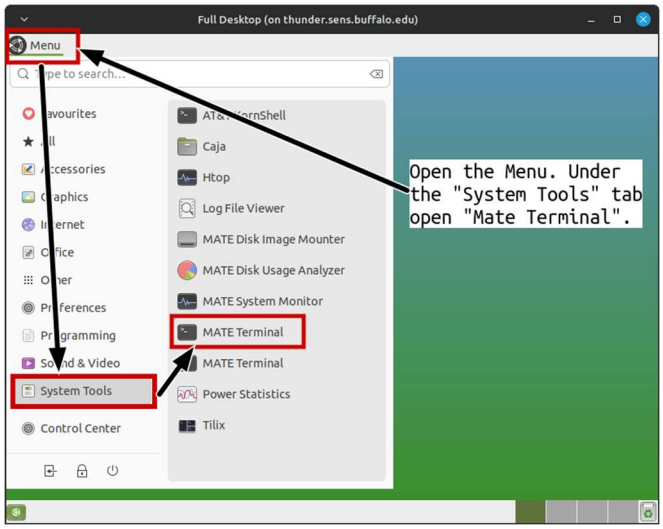
5. Once connected, select the "Full Desktop" option.



Step 2.2: Connect to CSE's "metallica" server

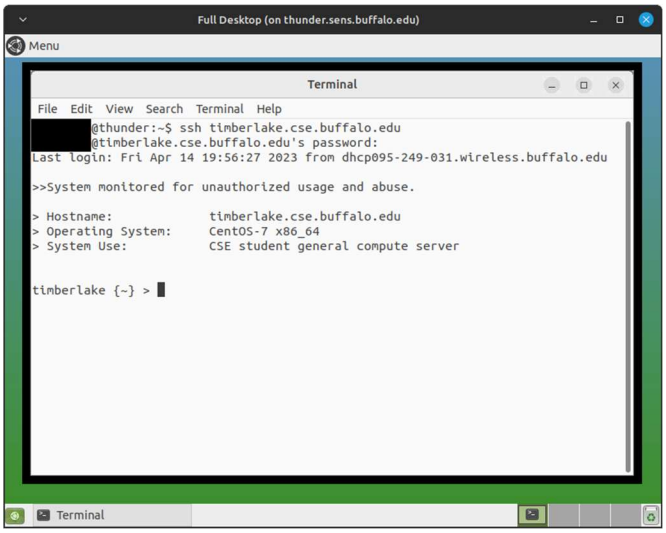
Perform the following INSIDE the virtual SENS desktop:

1. Open a terminal (inside the virtual SENS desktop, NOT directly on your personal computer).



2. Login to CSE's "metallica" server using the following command:

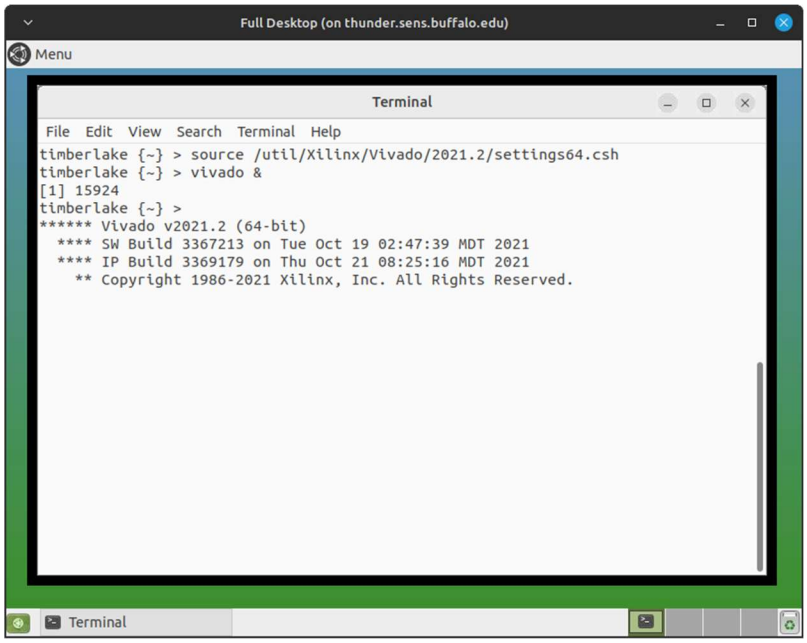
```
ssh metallica.cse.buffalo.edu
```



Step 2.3: Run Vivado Remotely on metallica

Inside the virtual SENS desktop, inside the terminal connected to metallica run the following commands to run Vivado:

```
source /util/Xilinx/Vivado/2021.2/settings64.csh
vivado &
```



3 – Steps for Remote Access using MacOS

1. Install XQuartz and restart your MacBook (you need to restart; the application needs to set up an environment variable on startup.)

2. Open Terminal, run (replacing [ubitname] with your username)

```
ssh [ubitname]@metallica.cse.buffalo.edu -X
```

3. Type "yes" when prompted to trust the connection, and login with your UBIT password.

4. Use the following commands to start Vivado:

```
source /util/Xilinx/Vivado/2021.2/settings64.csh
vivado &
```